Plendl Hourglass Model - FRED edition, run 27 October 2026

Model version 1.0.0. Data: FRED (Federal Reserve Bank of St. Louis), downloaded 2026-10-27. Trend lines fitted over FY2000-FY2026. Projection starts 01 October 2026, monthly, 40 years.

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Headline

• Starting debt D0: $40.463 T (Treasury Debt to the Penny, 2026-10-23); FY2026 year-end debt $40.172 T

• FY2026 receipts: $5.403 T; net interest: $1.044 T (estimated); primary deficit: $+0.948 T

• FY2026 interest-to-revenue (I/R): 19.3%; effective rate r0: 2.684%

• Currency in circulation: $2.416 T (2025-10-01), +2.6% year on year

• Observed NIPA interest / receipts (2026-04-01): 21.1% ($1.280 T / $6.056 T SAAR)

Calculated paradigm-shift point

Case	Paradigm-shift point	First month I/R >= 30%	First month I/R >= 40%
Case A: all trend lines fitted 2000-2026	None through September 2066	none	none
Case B: rate held at FY2026 level (dr = 0)	None through September 2066	Jan 2048	Sep 2059
Case C: rate slope fitted 2021-2026 only (+0.306 pp/yr)	June 2036 (I/R 55.4%)	May 2030	Mar 2033

Test: g_q(t) = 12 ln(q_t+1/q_t) and a(t) = g_q(t) - g_q(t-1). The paradigm-shift point is the first month from which, for every remaining month of the horizon, both g_q > 0 and a > 0. 30%/40% are reference lines only.

Fitted trend lines (OLS)

Series	Fit	Parameter	R^2
Receipts	log-linear	g_R = 4.21% / yr	0.917
Gross debt (reference)	log-linear	8.07% / yr	0.991
Effective rate (net interest / avg debt)	linear	dr = -0.0445 pp / yr	0.223
Primary deficit / receipts	linear	b_s = +1.221 pp / yr (start s0 = 17.6%)	0.174

Calculated values, yearly points (Case A; I/R for all cases)

Point	Debt $T	Interest $T	Revenue $T	Primary deficit $T	Rate %	I/R A %	I/R B %	I/R C %
Oct 2026	40.46	1.09	5.40	+0.95	2.68	20.1	20.1	20.1

Point	Debt $T	Interest $T	Revenue $T	Primary deficit $T	Rate %	I/R A %	I/R B %	I/R C %
Oct 2027	42.56	1.12	5.63	+1.06	2.64	20.0	20.3	22.6
Oct 2028	44.81	1.16	5.87	+1.17	2.59	19.8	20.5	25.3
Oct 2029	47.23	1.20	6.11	+1.30	2.55	19.7	20.8	28.2
Oct 2030	49.81	1.25	6.37	+1.43	2.51	19.6	21.0	31.4
Oct 2031	52.57	1.29	6.64	+1.57	2.46	19.5	21.4	34.8
Oct 2032	55.52	1.34	6.92	+1.72	2.42	19.4	21.7	38.5
Oct 2033	58.68	1.39	7.21	+1.88	2.37	19.3	22.1	42.5
Oct 2034	62.06	1.44	7.51	+2.05	2.33	19.2	22.5	47.0
Oct 2035	65.66	1.50	7.83	+2.23	2.28	19.2	22.9	51.9
Oct 2036	69.51	1.56	8.16	+2.43	2.24	19.1	23.3	57.3
Oct 2037	73.62	1.62	8.50	+2.63	2.19	19.0	23.8	63.2
Oct 2038	77.99	1.68	8.86	+2.85	2.15	18.9	24.3	69.8
Oct 2039	82.66	1.74	9.23	+3.09	2.11	18.9	24.8	77.1
Oct 2040	87.63	1.81	9.62	+3.33	2.06	18.8	25.4	85.2
Oct 2041	92.92	1.87	10.02	+3.60	2.02	18.7	25.9	94.3
Oct 2042	98.54	1.94	10.45	+3.88	1.97	18.6	26.5	104.3
Oct 2043	104.53	2.01	10.89	+4.17	1.93	18.5	27.2	115.5
Oct 2044	110.89	2.09	11.34	+4.49	1.88	18.4	27.8	127.9
Oct 2045	117.65	2.16	11.82	+4.82	1.84	18.3	28.5	141.8
Oct 2046	124.83	2.24	12.32	+5.17	1.79	18.2	29.2	157.4

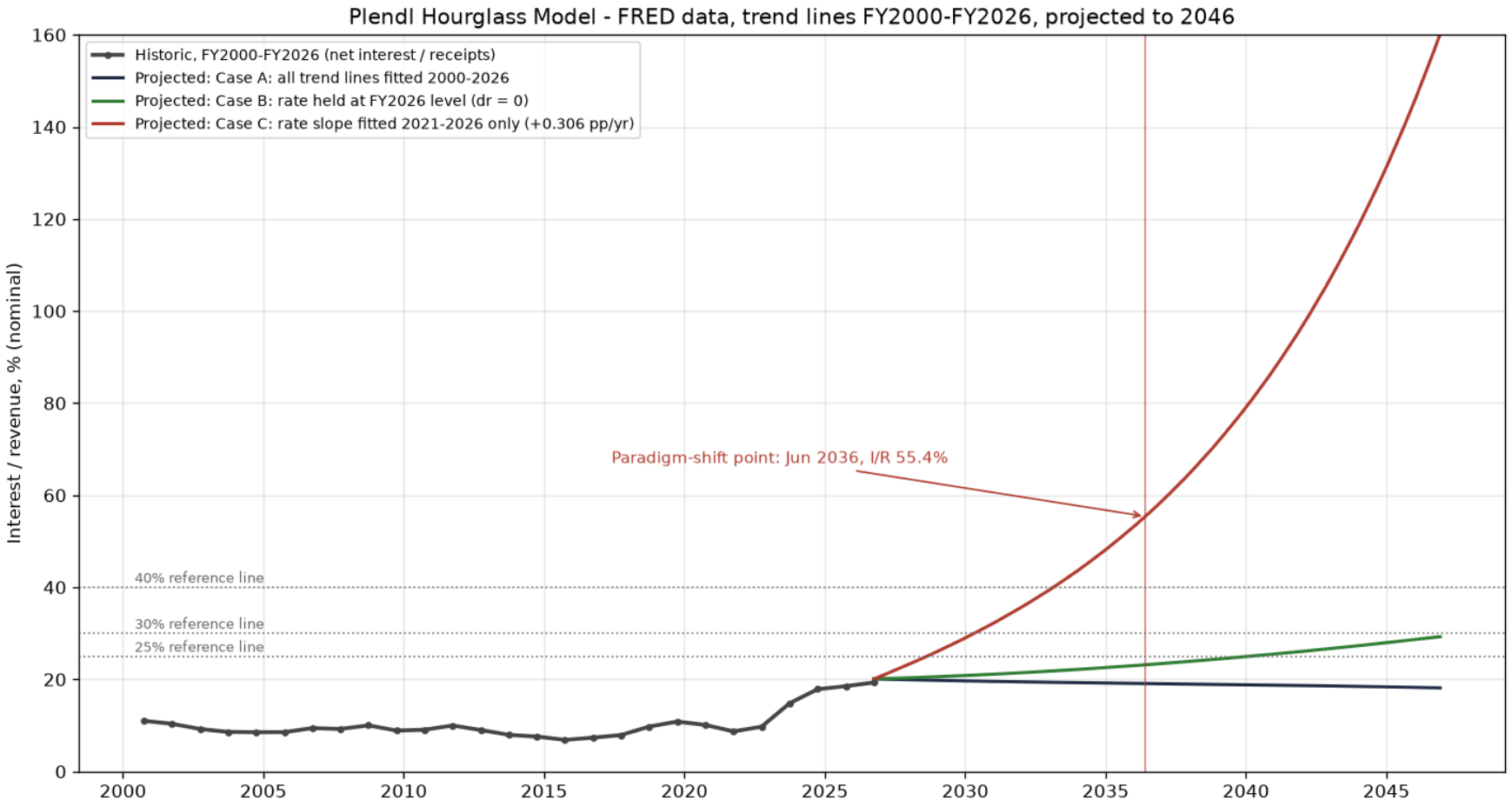
Historic data (FRED)

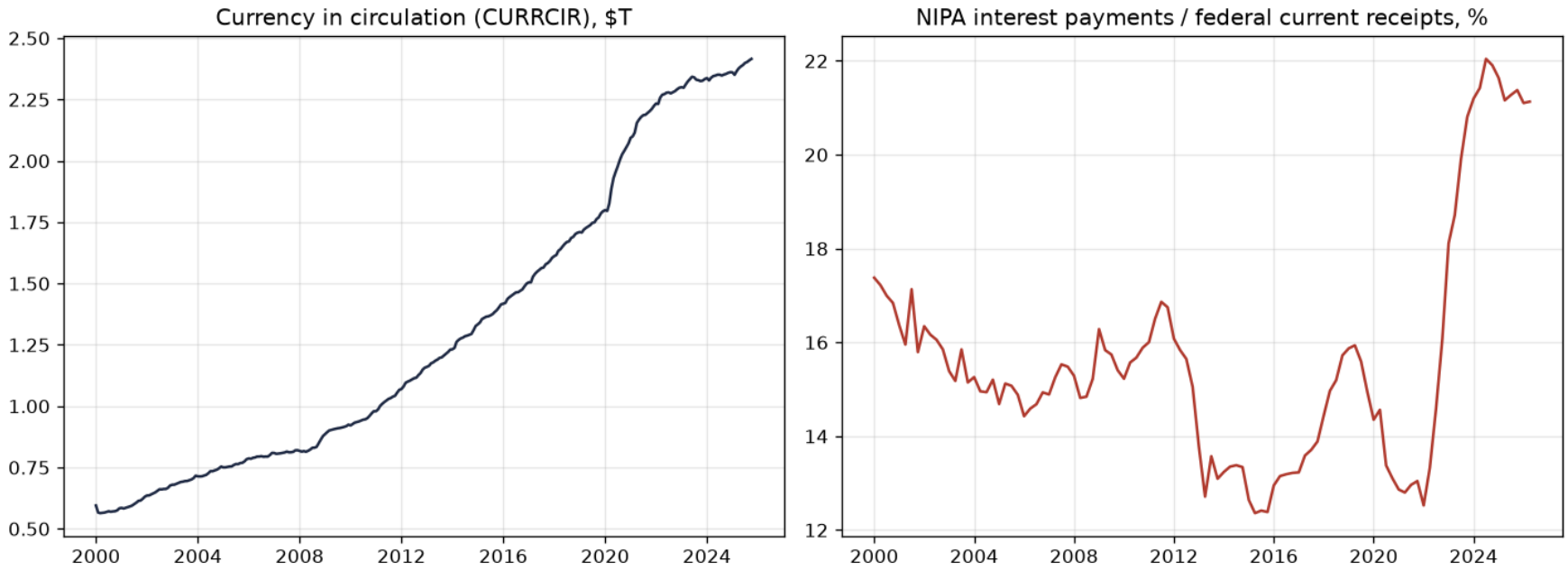
FY	Receipts $T	Outlays $T	Net interest $T	Primary deficit $T	Gross debt $T	Eff. rate %	I/R %
2000	2.025	1.789	0.223	-0.459	5.67	3.935	11.0
2001	1.991	1.863	0.206	-0.334	5.81	3.591	10.4
2002	1.853	2.011	0.171	-0.013	6.23	2.841	9.2
2003	1.782	2.160	0.153	+0.225	6.78	2.353	8.6
2004	1.880	2.293	0.160	+0.252	7.38	2.263	8.5
2005	2.154	2.472	0.184	+0.134	7.93	2.403	8.5
2006	2.407	2.655	0.227	+0.022	8.51	2.757	9.4
2007	2.568	2.729	0.237	-0.076	9.01	2.708	9.2
2008	2.524	2.983	0.253	+0.206	10.02	2.656	10.0
2009	2.105	3.518	0.187	+1.226	11.91	1.704	8.9
2010	2.163	3.457	0.196	+1.098	13.56	1.541	9.1
2011	2.303	3.603	0.230	+1.070	14.79	1.622	10.0
2012	2.450	3.527	0.220	+0.856	16.07	1.429	9.0

FY	Receipts $T	Outlays $T	Net interest $T	Primary deficit $T	Gross debt $T	Eff. rate %	I/R %
2013	2.775	3.455	0.221	+0.459	16.74	1.347	8.0
2014	3.021	3.506	0.229	+0.256	17.82	1.325	7.6
2015	3.250	3.692	0.223	+0.219	18.15	1.241	6.9
2016	3.268	3.853	0.240	+0.345	19.57	1.273	7.3
2017	3.316	3.982	0.263	+0.403	20.24	1.319	7.9
2018	3.330	4.109	0.325	+0.454	21.52	1.556	9.8
2019	3.463	4.447	0.375	+0.608	22.72	1.696	10.8
2020	3.421	6.517	0.345	+2.750	26.95	1.391	10.1
2021	4.047	6.821	0.352	+2.421	28.43	1.273	8.7
2022	4.897	6.272	0.476	+0.898	30.93	1.603	9.7
2023	4.441	6.128	0.658	+1.029	33.17	2.054	14.8
2024	4.920	6.735	0.880	+0.935	35.46	2.564	17.9
2025	5.236	7.011	0.970	+0.805	37.64	2.654	18.5
2026	5.403	7.396	1.044*	+0.948	40.17	2.684	19.3

Note: FY2026 receipts and outlays are actuals (sum of the 12 monthly MTS values, Oct 2025-Sep 2026). FY2026 net interest (*) is an ESTIMATE until FRED's FYOINT posts FY2026: FY2025 x 1.0763 (growth of NIPA interest payments, last 4 vs prior 4 quarters, through 2026-04-01).

Note: Debt (all years) = Treasury Debt to the Penny at fiscal-year end; FY2026: $40.172 T (2026-09-30). Starting debt D0 = latest Debt to the Penny, $40.463 T (2026-10-23). For reference, FRED GFDEBTN latest = $39.462 T (2026-04-01, quarter start date) and FYGFD FY2025 = $37.375 T (OMB concept).


Plendl Hourglass Model - FRED data, trend lines FY2000-FY2026, projected to 2046
Historic, FY2000-FY2026 (net interest / receipts)
Projected: Case A: all trend lines fitted 2000-2026
Projected: Case B: rate held at FY2026 level (dr = 0)
Projected: Case C: rate slope fitted 2021-2026 only (+0.306 pp/yr)
Interest / revenue, % (nominal)
Paradigm-shift point: Jun 2036, I/R 55.4%
40% reference line
30% reference line
25% reference line
0
20
40
60
80
100
120
140
160
2000
2005
2010
2015
2020
2025
2030
2035
2040
2045



Data series

- FYFR: Federal Receipts, fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYFR
- FYONET: Federal Net Outlays, fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYONET
- FYOINT: Federal Outlays: Interest, fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYOINT
- FYGFD: Gross Federal Debt, fiscal year end (reference only) (billions $, annual FY) - https://fred.stlouisfed.org/series/FYGFD
- GFDEBTN: Federal Debt: Total Public Debt (reference only) (millions $, quarterly) - https://fred.stlouisfed.org/series/GFDEBTN
- MTSR133FMS: Monthly Treasury Statement: Total Receipts (millions $, monthly) - https://fred.stlouisfed.org/series/MTSR133FMS
- MTSO133FMS: Monthly Treasury Statement: Total Outlays (millions $, monthly) - https://fred.stlouisfed.org/series/MTSO133FMS
- A091RC1Q027SBEA: Federal government current expenditures: Interest payments (billions $ SAAR, quarterly) - https://fred.stlouisfed.org/series/A091RC1Q027SBEA
- FGRECPT: Federal government current receipts (billions $ SAAR, quarterly) - https://fred.stlouisfed.org/series/FGRECPT
- CURRCIR: Currency in Circulation (billions $, monthly) - https://fred.stlouisfed.org/series/CURRCIR
- Treasury Debt to the Penny (total public debt outstanding, daily, $) - https://api.fiscaldata.treasury.gov/services/api/fiscal_service/v2/accounting/od/debt_to_penny